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Comparison of geometrical parameters of rotary module for modular construction machinery¹

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Abstract. Optimization of curvature angle of rotating module is dealt with. The angle of curvature has a significant impact on the size of the workspace of modular kinematic structure constructed from such modules. The aim is to find the angle of curvature at which the maximum possible range is achieved and structure does not collide with itself. In case of comparison of geometric parameters of rotary module for modular construction machinery a mathematical model was used with utilization of Denavit–Hartenberg principle of coordinate systems deployment.

Key words. Modular machine, rotation module, modular robot, machine workspace.

1. Introduction

The constructional layout of a rotation module (Fig. 1) protected by a utility model [1] can be used to build serial robotic structures (Fig. 2) and, in special cases, also production machines [2].

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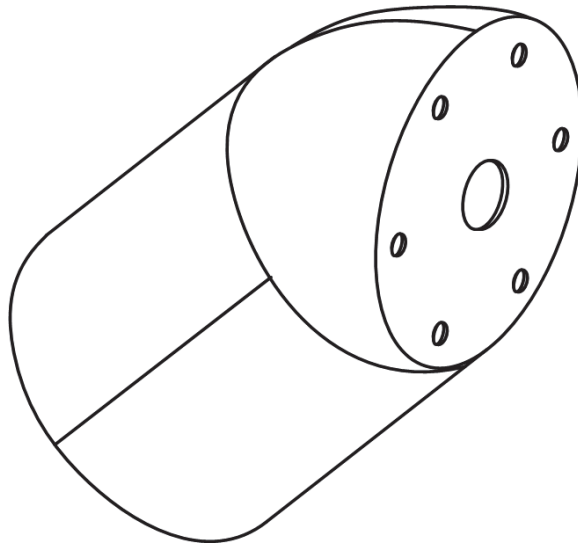


Fig. 1. Basic building unit-autonomous rotary module with 45° angle of curvature

rotating module

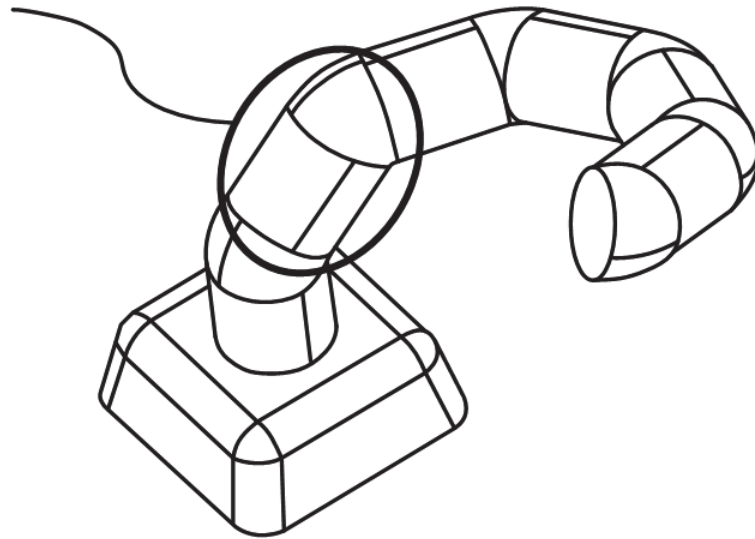


Fig. 2. Scheme of serial homogeneous kinematic structure with 6 DOFs

Every module is inside equipped with a DC servo motor with a three-way incremental encoder with 512 (default) positional sensings, reducer and controlled electromagnetic brake. At one end of each module there is a bayonet flange in the perpendicular direction to the axis of the engine and at the other end another flange in an angle adjusted to the curvature of the axis of the next module. Using this way of mutual coupling we can build serial kinematic structures with varying numbers of the degrees of freedom (DOFs). If an extension piece is inserted between the rotary modules, we can substantially affect the range of that serial kinematic structure and the shape of its workspace.

2. Mathematical description of the mechanism

The investigation of the kinematic chain consisting of a rotary motion unit requires a specific mathematical description. The most appropriate way for such a spatial mechanism is to apply the Denavit–Hartenberg (D–H) principle. This principle characterizes the arrangement of the coordinate systems of individual modules at the points of the rotational or translational motion. It starts from the assumption that any two coordinate systems can be described by the group of four variables θ , d , a , α , denoting an angle of rotation, a pair of translational shifts and next angle of rotation, respectively. The principle of location of the coordinate systems is known [3]. The basic transformation matrix between two coordinate systems located at the joints of motion units is

$$\mathbf{A}_{i-1}^i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (1)$$

For a homogeneous modular serial mechanism consisting of modules with angle of curvature $\alpha = \frac{1}{4}\pi$ and distances l_1 and l_2 between a pair of adjacent coordinate systems we obtain

$$\mathbf{A}_{i-1}^i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \frac{1}{4}\pi & \sin \theta_i \sin \frac{1}{4}\pi & l_2 \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \frac{1}{4}\pi & -\cos \theta_i \sin \frac{1}{4}\pi & l_2 \sin \theta_i \\ 0 & \sin \frac{1}{4}\pi & \cos \frac{1}{4}\pi & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2)$$

The angle θ_i represents a common variable, for which we will substitute different values from the specified interval.

In the next text we will present an analysis of a mechanism with 5 DOFs that consists of identical modules. The analysis was performed for the modules of curvatures corresponding to the angles 30° , 45° , 60° and 90° . In this case the total transformation matrix between the basic coordinate system LCS_b and LCS_5 (local coordinate system) has the form

$$\mathbf{T}_b^5(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5) = \mathbf{A}_b^0 \mathbf{A}_0^1(\theta_1) \mathbf{A}_1^2(\theta_2) \mathbf{A}_2^3(\theta_3) \mathbf{A}_3^4(\theta_4) \mathbf{A}_4^5(\theta_5). \quad (3)$$

Conversion of the coordinates of the last LCS_5 to the basic LCS_b is given by the formula

$$\begin{bmatrix} x_b \\ y_b \\ z_b \\ 1 \end{bmatrix} = \mathbf{T}_b^5(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5) \begin{bmatrix} x_5 \\ y_5 \\ z_5 \\ 1 \end{bmatrix}. \quad (4)$$

For the purposes of graphical and numerical calculation of specific values of points of the workspace of kinematic structures with 5 DOFs, the MathCAD

program was applied. The variables in the movement matrices (in this case the values of the angles of rotation of particular basic modules θ_i) were taken from intervals $(0, 2\pi)$ with a half-radian step. A finer division is possible, but it considerably increases the time of calculations, and it is of rather low importance with respect to rendering the work area.

3. Analysis of workspace

One of the most important parameters of building module (Fig. 1) is the angle of curvature. In linking these modules into a homogeneous serial kinematic chain with, e.g., 5 DOFs, we can build mechanisms with similar but also with partly different shapes. It is also interesting to compare the range of the kinematic mechanisms. From the results of analysis (Fig. 3) it is clear that the larger the angle of curvature is, the greater range is in almost every axis (Table 1).

The above rule, however, has an exception. When comparing the mechanisms with the angles of curvature 45° and 30° , for the z -axis we obtain a smaller range in the case of angle 45° . The influence of the angle of curvature on the workspace (for angles 30° , 45° , 60° and 90°) follows from Fig. 3.

Figure 3 does not contain the second view in plane x - y . This missing view for a homogeneous kinematic structure composed of modules with the angle of 30° is shown in Fig. 4. Due to a high similarity of the shape of workspace for other structures consisting of modules with different angles of curvature, other range values are only listed in Table 1. This table contains (besides the minimum and maximum values in each of the three axes x , y , z denoted as $x_{\min}$, $x_{\max}$, $y_{\min}$, $y_{\max}$, $z_{\min}$, $z_{\max}$) also the corresponding ranges defined as $r_x = x_{\max} - x_{\min}$, $r_y = y_{\max} - y_{\min}$ and $r_z = z_{\max} - z_{\min}$. The final three columns contain the values $xyz_{\min} = (x_{\min} + y_{\min} + z_{\min})/3$, $xyz_{\max} = (x_{\max} + y_{\max} + z_{\max})/3$ and $r_{xyz} = (r_x + r_y + r_z)/3$. The last value gives an idea about the absolute range of the kinematic structure and, usually, there holds that the greater is the value, the greater is the range. But it may not apply in all directions. It is only the averaged value of the absolute ranges.

For a complete verification of the workspace range it would be necessary to analyze it for finer gradations of the curvature angle.

As evident from Fig. 5, the largest range in all considered axes has a homogeneous serial kinematic structure with the maximum angle of curvature 90° . For the configuration of the structure with that curvature of the basic module, the danger of the self-collision has to be taken into account, as can be seen in Fig. 3 (space representation of the structure). The self-collision problem can be addressed by appropriate control or by software restrictions in the management program.

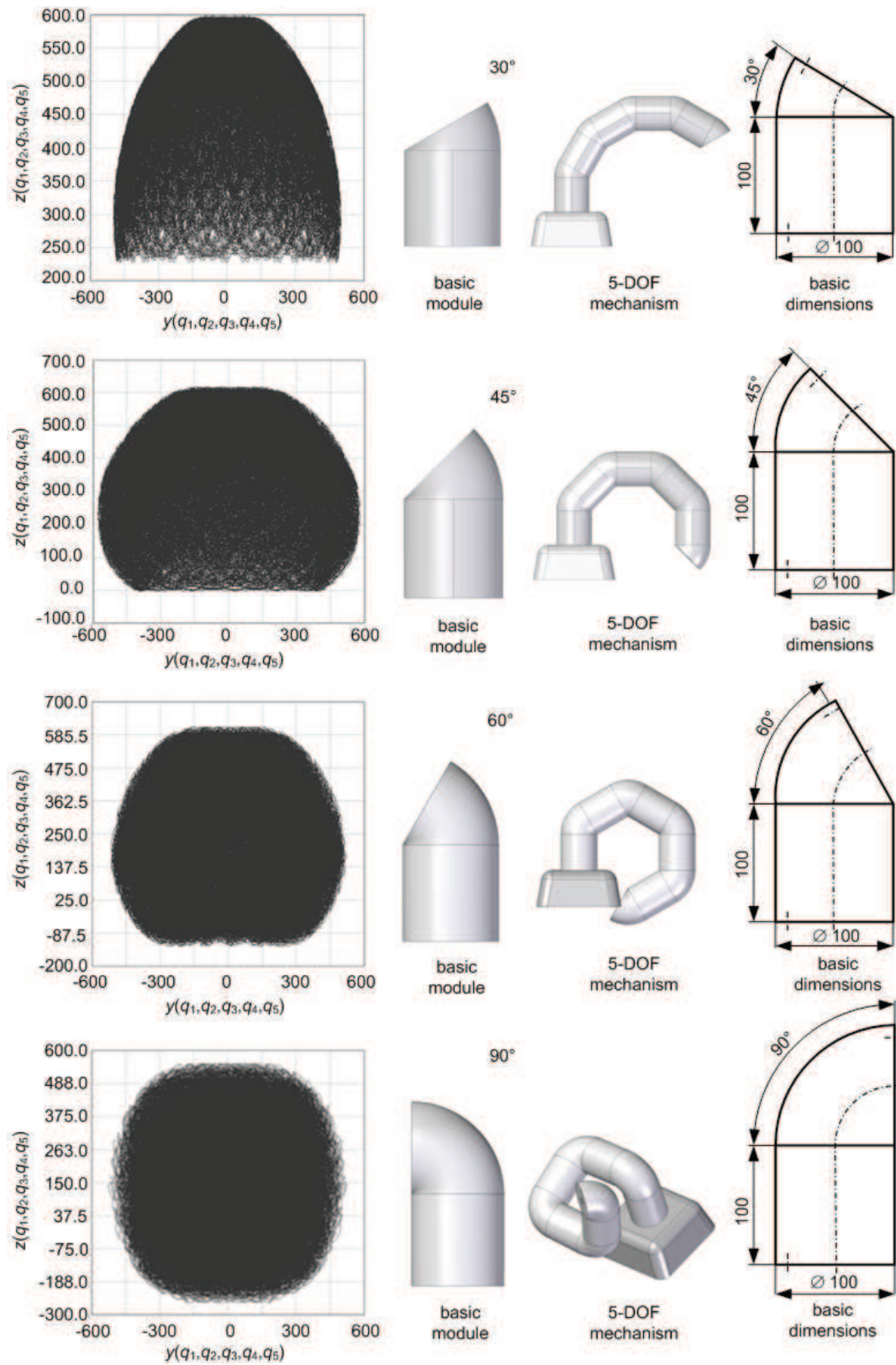


Fig. 3. Range of modular serial chain with 5 degrees of freedom and 30° curvature in x - y plane

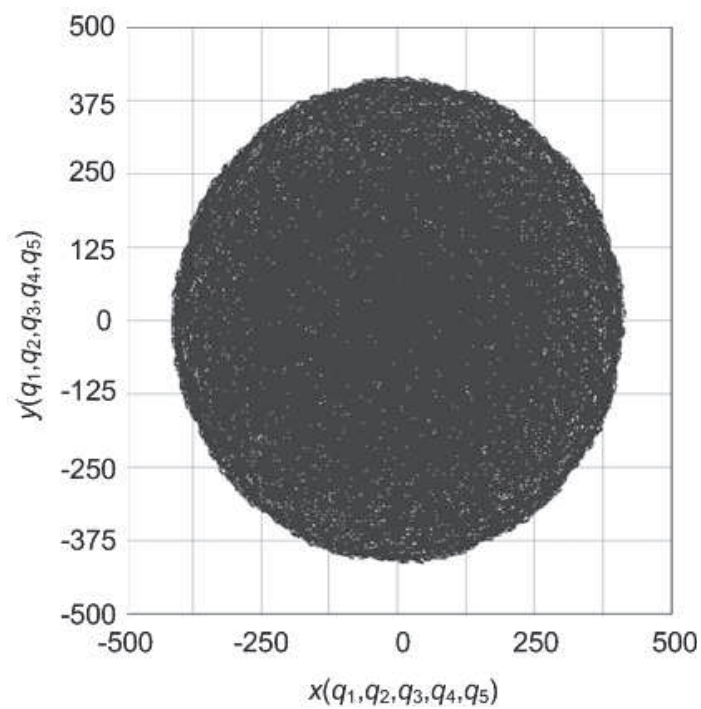


Fig. 4. Footprint (plane x – y) for serial homogeneous kinematic structures with 5 DOFs and curvature angle of module 30°

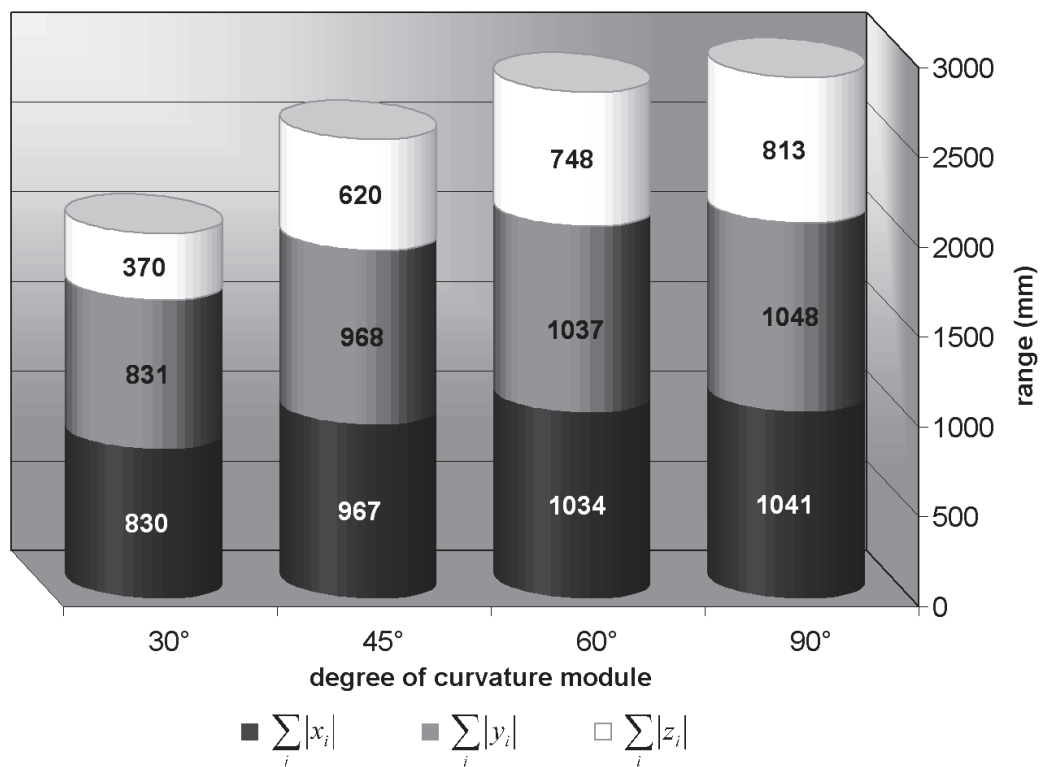


Fig. 5: Direct comparison of the range of kinematic structures consisting of modules with varying degrees of curvature (from 30° to 90°)

Table 1. Resulting values of the direct kinematics functions generated by MathCAD

angle of curvature	$x_{\min}$	$x_{\max}$	r_x	$y_{\min}$	$y_{\max}$	r_y	$z_{\min}$	$z_{\max}$	r_z	$xyz_{\min}$	$xyz_{\max}$	r_{xyz}
	(mm)											
30°	-414	416	830	-415	416	831	228	598	370	-200	477	677
45°	-483	484	967	-484	484	968	-3	617	620	-323	528	852
60°	-518	516	1034	-518	519	1037	-133	615	748	-390	550	940
90°	-525	516	1041	-527	521	1048	-258	555	813	-437	531	967

Another way of solving this problem can be offered for the structures with 5 DOFs for angles less than 60° (Fig. 3). For the structure with more DOFs, such angles have to be reduced so that their sum on the basic module is no more than 240°.

4. Conclusion

There exist many approaches to solution and analysis of technical systems, using the simulation system in the process of experimental measurement [4]. The analysis of the range parameters of a serial homogeneous kinematic structure and its virtual model represent the first step toward a study of this new kinematic arrangement. The evaluation of the first results of the simulation shows that further study of the structure is desirable. The main advantage of the structure is its design simplicity, excellent spatial range, variability and flexibility of modular combinations and, thus, many options of its arrangement. Further developments assume the unification of rotary modules and their integration into the design of mutually connectable modular series.

To transmit data concerning the structure, wireless technologies are assumed. This variant will provide successful solution on motion limitation and rotation of particular basic modules, and it will lead to elimination of restriction typical for solutions that are currently offered on the worldmarket.

The presented constructional design of rotating modules is intended for the robotic manipulators with less dynamic motion and after suitable adaptations also for various production machines with special functions.

For the implementation of prototype, its successful testing and future application in practice, it is also necessary to analyse the potential risks [5].

The presented principle is currently in the process of logging in to the Office of Patent Ownership of the Slovak Republic as a utility model [1]. It is closely related to the patent proposal [6].

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